Validation of Control Regions for a Data-Based Background Estimate for a Dark Photon Search at CMS

Claire O'Brien-Dull, Malcolm Wilson-Ahlstrom, Marcus Kubon, Gavin O'Malley,
Sreevardhan Atyam, Vikram Rao, Albert Han, Ivan Chen, Catherine Jenks, Peter Dong,
Lenny Speigel, Paul Karchin

The Dark Photon Model:

In an attempt to solve the dark matter problem, some propose a dark sector. This dark sector includes a dark matter fermion χ and a massive U(1)' gauge boson A' (dark photon). The main production mechanism for the dark photon would be an s-channel resonance through a non-SM Higgs boson with a baseline mass of 1000 GeV, which could couple to both SM quarks and dark matter particles. A' decays into SM particles through kinetic mixing. The high mass of the non-SM Higgs causes a large Lorentz in the resulting leptons, collimating them into a lepton jet. This is easily recognizable at the LHC, and we are therefore left with a method of A' detection, despite its lack of direct interaction with SM particles.

Lepton Jet Selection:

We use only muons in Lepton Jet reconstruction. We have found that electrons do a poor job of reconstructing lepton jets. We think this is because they are too close together for us to distinguish them and making it difficult to find both electrons in each electron pair.

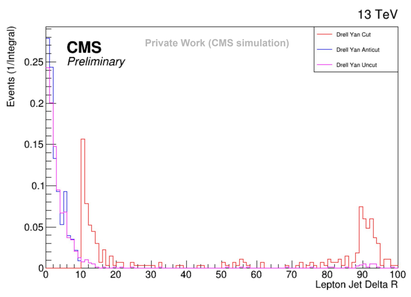
Two Control Regions:

We took two control regions that are kinematically similar to our signal but do not pass all our cuts: high mass and displaced vertex. Here, we tested whether our proposed control regions would be effective by using Monte Carlo to compare inside and outside control regions. Now, we can apply the same control regions to our data to get a background estimate. This is useful because rates of lepton jet fakes are hard to model in Monte Carlo, so we can use a data-driven background to calculate them in control regions. Our next step would be to take our data from the control region and extrapolate it to our signal to find how much background we really have.

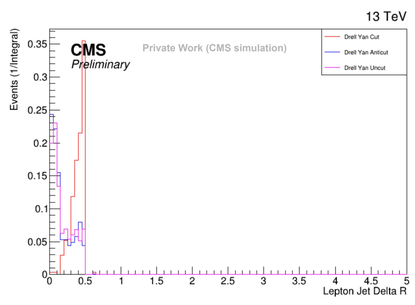
High mass:

We set the mass of the dark photon to be from 0.1 to 4.0 GeV because this region is the best for collider detection using lepton jets, as higher masses don't form lepton jets as easily. However, lepton jets are also formed from Z decays - a background process much more massive than our signal. Our signal region requires <10 GeV, so we use the region about 10 GeV as a control region.

We saw little difference in eta and phi between low-mass and high-mass regions but large difference in pt and delta r between low-mass and high-mass regions .



Lepton Jet Pt

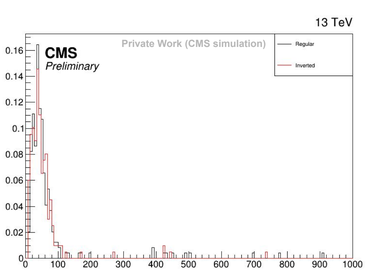


Lepton Jet ΔR

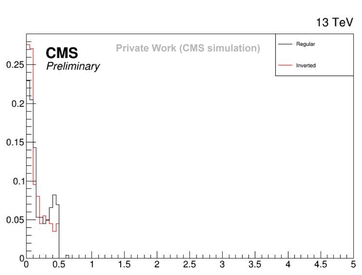
Displaced vertex:

Our signal comes from the primary vertex, so we can examine lepton jets that are further away from the primary vertex as a control region. The cut is dXY>0.0001 and dZ>0.0005, creating our control region.

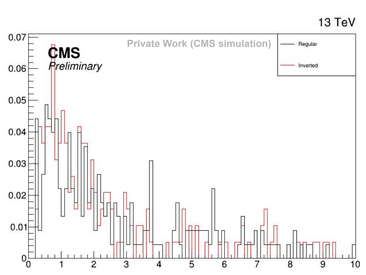
Our inverted and regular plots are similar, which proves that they could be effectively used as a control region. However, our control region makes it impossible to reconstruct dark photons that have a slightly longer lifetime so that they have displaced vertices that are still within the beam pipe.



Background Pt

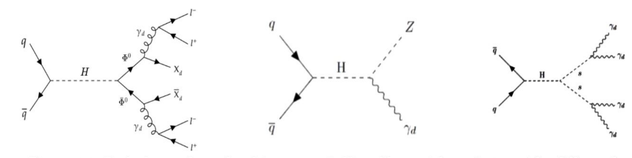


Background ΔR

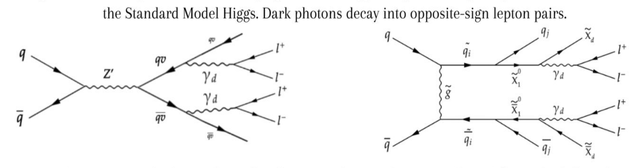


Background Mass

Production Portals:



Figures 1-3: Dark photon channels relying on exotic Higgs-like particles, spin 0 particles different from the Standard Model Higgs. Dark photons decay into opposite-sign lepton pairs.

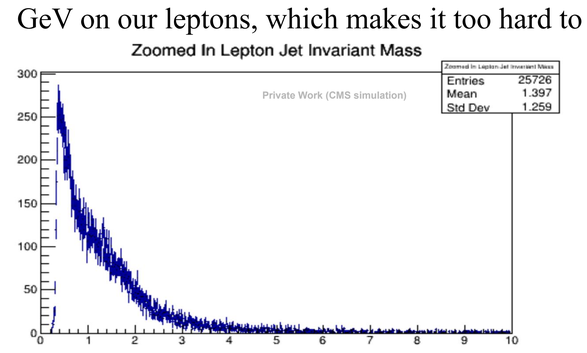


Figures 4-5: Dark photon channels relying on other mechanisms: a heavy Z' particle (left) and supersymmetric particles (right)

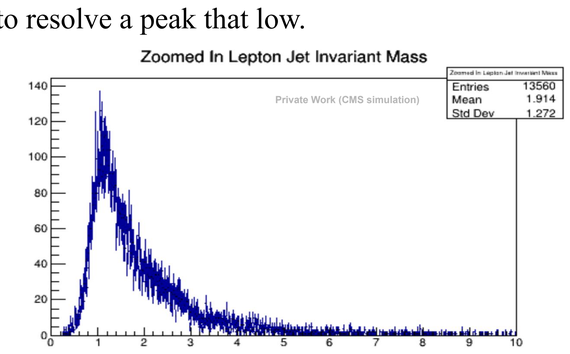
We found that the kinematics of the lepton jet do not depend on the production mechanism, as long as they come from a massive particle. On the left are five possible theoretical mechanisms we use. The top left mechanism is our baseline portal, but we use the others to show we have a mechanism-independent result.

Lepton Jet Invariant Mass Histograms:

We reconstructed lepton jets from dark photon events, and created histograms to model the invariant mass of the histogram. The cone size is 0.01. These histograms prove we are reconstructing the lepton jets effectively, but our reconstruction seems to fail around 0.5 GeV, which can be observed in the 0.3GeV histogram. This is because of the pT cut of 5 GeV on our leptons, which makes it too hard to resolve a peak that low.



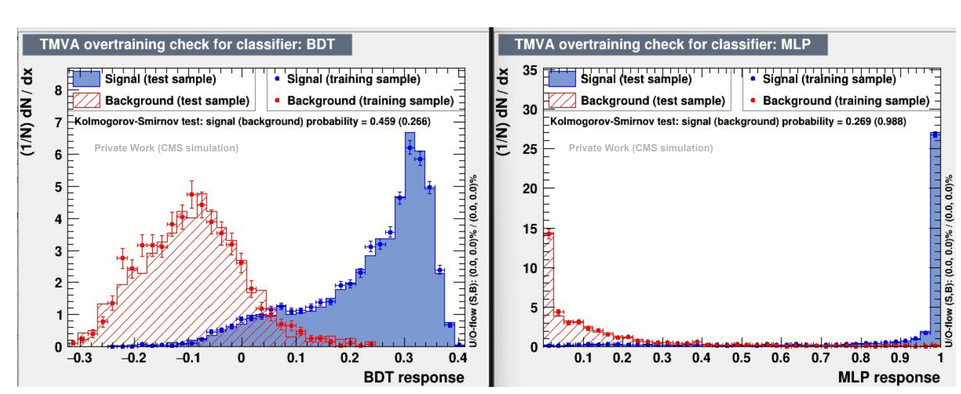
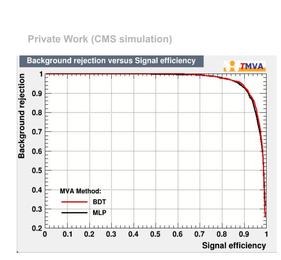
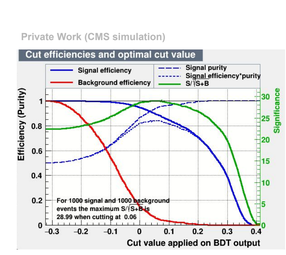
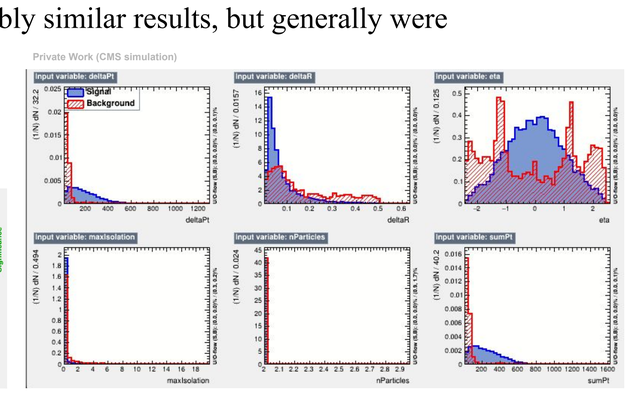
Dark Photon Mass 0.3 GeV



Dark Photon Mass 1.2 GeV

Neural Network Methodology

Dark photons can interact with Standard Model particles through kinetic mixing, allowing them to decay into Standard Model leptons, from which lepton jets are produced. However, classification of such particles becomes challenging when introducing background effects, systematic errors, and incorrect cut values. Therefore, we tested a novel methodology in which a multivariate analysis is used to produce a neural network that determines optimal cut efficiencies for dark photon recognition.



Past experiment results that reflect similar fake jet ratios for both background and signal were refuted, but low overall background representation of ~20%, supports that multi-flavor lepton jets are not a good candidate for a control sample or fake estimate. Moreover, several neural net methods for cut efficiency and optimal cut values were employed: Boosted Decision Trees and Multi-Layer Perceptrons. We finalized on using number of particles, isolation, eta, cone size, and total/change in pTas variables due to their high of signal-background separation, noted by signal peaks. Through BDT efficiency ROC curves, we concluded that the sensitivity for our experiment was maximized at 28.99, with around 93% background rejection and signal efficiency . MLPs also had remarkably similar results, but generally were outperformed by BDTs.